

Day - 02

Bits, Qubits and Quantum States

Before diving deep into quantum computing, we must fundamentally understand how information is stored and represented. Classical computers use **Bits**, while quantum computers use **Qubits**. The way these two systems process information represents a massive paradigm shift.

1. Classical Information: Bits

A **bit** is the smallest unit of information in a classical computer. It can exist in only one of two definite states at any given moment: **0 (OFF)** or **1 (ON)**.

- **Analogy:** Think of a light switch. It is either completely ON or completely OFF. It cannot be both at the same time.
- **Power of Scaling:** Computers combine multiple bits to represent larger amounts of information. The number of possible states grows exponentially as 2^n , where n is the number of bits.

Number of Bits	Possible Values
1 Bit	2 Values (0, 1)
2 Bits	4 Values (00, 01, 10, 11)
3 Bits	8 Values
8 Bits	256 Values
10 Bits	1024 Values

2. Quantum Information: Qubits

A **qubit** (quantum bit) is the fundamental unit of information in a quantum computer. Unlike a classical bit, a qubit is not restricted to just 0 or 1. Due to the principles of quantum mechanics, a qubit can exist in a **combination** of both states simultaneously.

The Spinning Coin Analogy

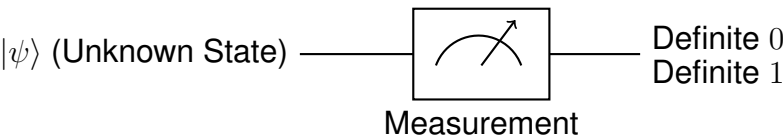
Imagine a coin. When it is resting on a table, it is clearly Heads (1) or Tails (0)—this is like a classical bit. However, when the coin is **spinning**, it is a blur of both Heads and Tails. Before it lands, it is not fixed to one outcome. This spinning state is analogous to a qubit in **superposition**.

Real Hardware Implementations: Qubits are physical systems, not just mathematical concepts. Modern quantum computers (like IBM's) create qubits using superconducting circuits cooled to near absolute zero, trapped ions, photons, or neutral atoms.

3. Quantum States & Measurement

A **quantum state** describes the complete condition of a qubit before it is measured. It stores the **probability** of obtaining a specific result.

- **Ket Notation:** In quantum computing, we represent the basic states using Dirac notation (kets): $|0\rangle$ and $|1\rangle$.
- **Superposition:** A qubit can exist in a state where there is, for example, a 50% chance of measuring 0 and a 50% chance of measuring 1.



State Collapse: The moment you observe (measure) a qubit in a superposition, its state **collapses**. The original rich quantum state is destroyed, leaving you with a definite, classical result of either 0 or 1.

4. Bits vs. Qubits: A Summary Comparison

Feature	Classical Bit	Quantum Qubit
Values	0 or 1	0, 1, or a combination
Underlying Physics	Classical Physics	Quantum Mechanics
State Before Observation	Fixed and definite	Probabilistic
Information Capacity	Limited (one value)	Richer Representation
Effect of Measurement	No Change	State Collapse

End-of-Day Key Takeaways

- **Bits** represent definitive, binary states (0 or 1) and form the foundation of all classical computing.
- **Qubits** harness quantum mechanics to exist in **superposition**, allowing quantum computers to explore massive solution spaces simultaneously.
- **Measurement** permanently forces a probabilistic quantum state to collapse into a definite classical state.